

Predictive Modeling

A Project Report

Submitted to



By

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Context:

An over-the-top (OTT) media service is a media service offered directly to viewers via the internet. The term is most synonymous with subscription-based video-on-demand services that offer access to film and television content, including existing series acquired from other producers, as well as original content produced specifically for the service. They are typically accessed via websites on personal computers, apps on smartphones and tablets, or televisions with integrated Smart TV platforms.

Presently, OTT services are at a relatively nascent stage and are widely accepted as a trending technology across the globe. With the increasing change in customers' social behavior, which is shifting from traditional subscriptions to broadcasting services and OTT on-demand video and music subscriptions every year, OTT streaming is expected to grow at a very fast pace. The global OTT market size was valued at

121.61 billion by 2019 and projected to reach 1,039.03 billion by 2027, growing at a CAGR of 29.4% from 2020 to 2027. The shift from television to OTT services for entertainment is driven by benefits such as on-demand services, ease of access, and access to better networks and digital connectivity.

With the outbreak of COVID-19, OTT services are striving to meet the growing entertainment appetite of viewers, with some platforms already experiencing a 46% increase in consumption and subscriber count as viewers seek fresh content. With innovations and advanced transformations, which will enable the customers to access everything they want in a single space, OTT platforms across the world are expected to increasingly attract subscribers on a concurrent basis.

Objective:

ShowTime is an OTT service provider and offers a wide variety of content (movies, web shows, etc.) for its users. They want to determine the driver variables for first-day content viewership so that they can take necessary measures to improve the viewership of the content on their platform. Some of the reasons for the decline in viewership of content would be the decline in the number of people coming to the platform, decreased marketing spend, content timing clashes, weekends and holidays, etc. They have hired you as a Data Scientist, shared the data of the current content on their platform, and asked you to analyze the data and come up with a linear regression model to determine the driving factors for first-day viewership.

Data Description:

The data contains the different factors to analyze for the content. The detailed data dictionary is given below.

Data Dictionary:

Variable	Description
visitors	Average number of visitors, in millions, to the platform in the past week
ad_impressions	Number of ad impressions, in millions, across all ad campaigns for the content (running and completed)
major_sports_event	Any major sports event on the day
genre	Genre of the content

dayofweek	Day of the release of the content
season	Season of the release of the content
views_trailer	Number of views, in millions, of the content trailer
views_content	Number of first-day views, in millions, of the content

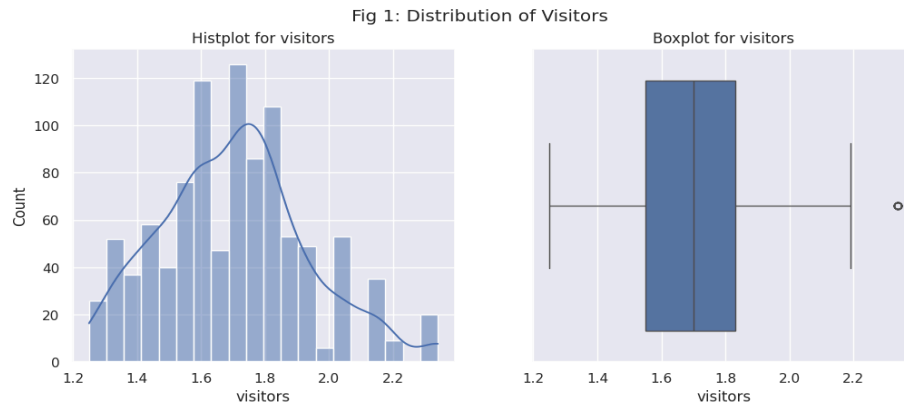
Data Overview

1. Import the required packages
2. Loaded the ott dataset using **read_csv()**.
3. Used the **head()** to get the first five rows of data.
4. Checked the shape of the dataset using the **shape method**.
 - a. The number of rows is 1000.
 - b. The number of columns is 8.
5. Checked the data types of all the columns in the dataset using **info()**.
 - a. The visitors, ad_impressions, views_trailer and views_content belong to float type.
 - b. The major_sports_event belongs to int type.
 - c. The genre, dayofweek and season belongs to object type.
 - d. There are no missing values in the dataset.
6. Checked the statistical summary of the given data set using **describe()**.
 - a. In the genre column the number of unique values is 8 and the most repeated value is Others(255).
 - b. In the dayofweek column the number of unique values are 7 and the most repeated value is Friday(369).
 - c. In the season column the number of unique values are 4 and the most repeated value is Winter(257).

Exploratory Data Analysis

Univariate Analysis

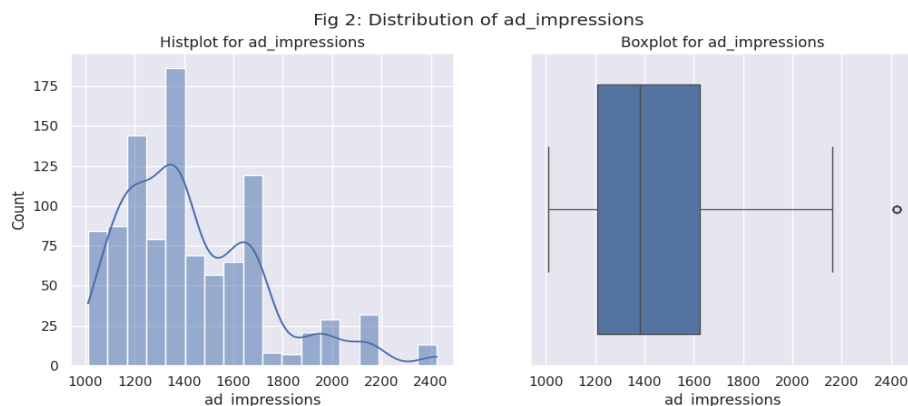
Understanding the distribution of visitors



Observations:

1. Almost a normal distribution with only a single outlier can be observed.
2. Mean and median follows almost similarly at 1.7 million visitors.

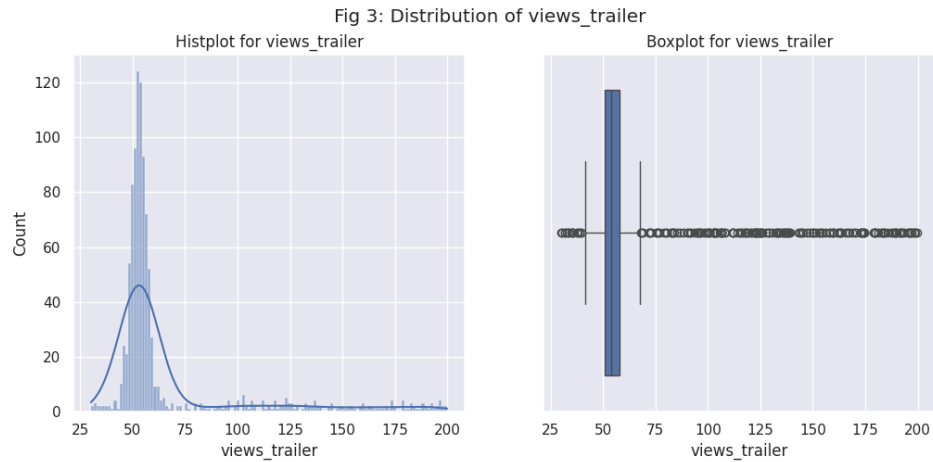
Understanding the distribution of ad_impressions



Observation:

1. Less than 0.5 million watched the actual content.
2. Many outliers can be observed. Distribution is right skewed. However, it seems like a normal distribution.

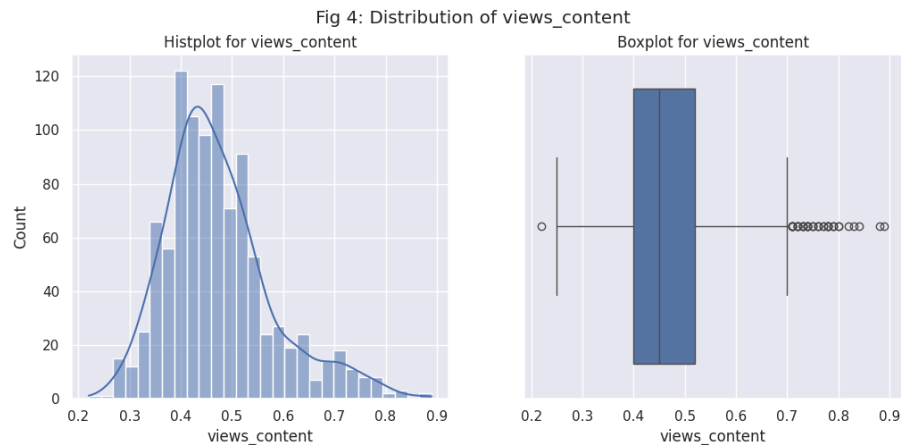
Understanding the distribution of views_trailer



Observation:

1. Too many outliers can be observed at both sides.
2. Heavily right-skewed data.
3. The Mean is less than 70 million viewed the trailer.

Understanding the distribution of views_content



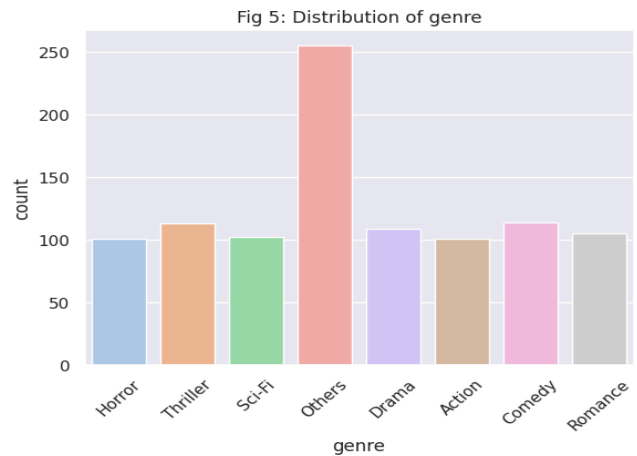
Observation:

1. Less than 0.5 million watched the actual content.
2. Many outliers can be observed.
3. Distribution is right skewed. However, it seems like a normal distribution.

Understanding the distribution of genre

Observation:

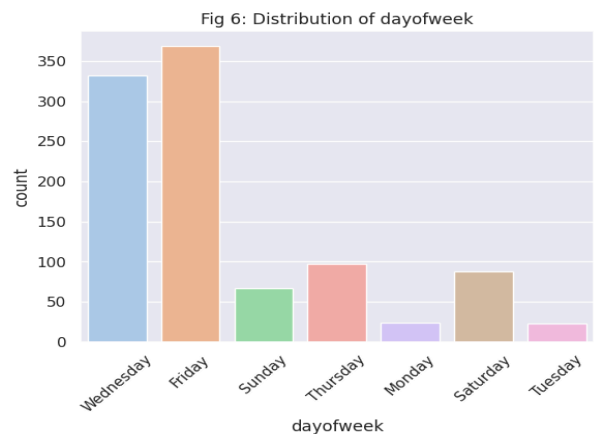
1. Genre “others” has the maximum counts in the data followed by Comedy & Thriller.



Understanding the distribution of dayofweek

Observation:

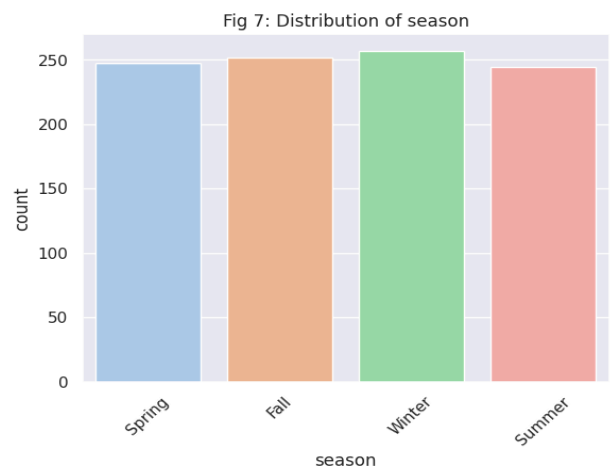
1. Mostly content released on Friday followed by Wednesday.
2. Very less content released on Monday.



Understanding the distribution of season

Observation:

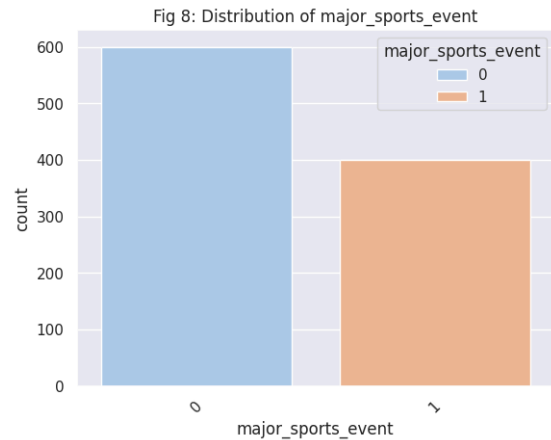
1. There are four seasons in the data.
2. Mostly content released in Winter.
3. The remaining seasons have almost equal values.



Understanding the distribution of major_sports_event

Observation:

1. Mostly the content released when there is no major sport event happening.

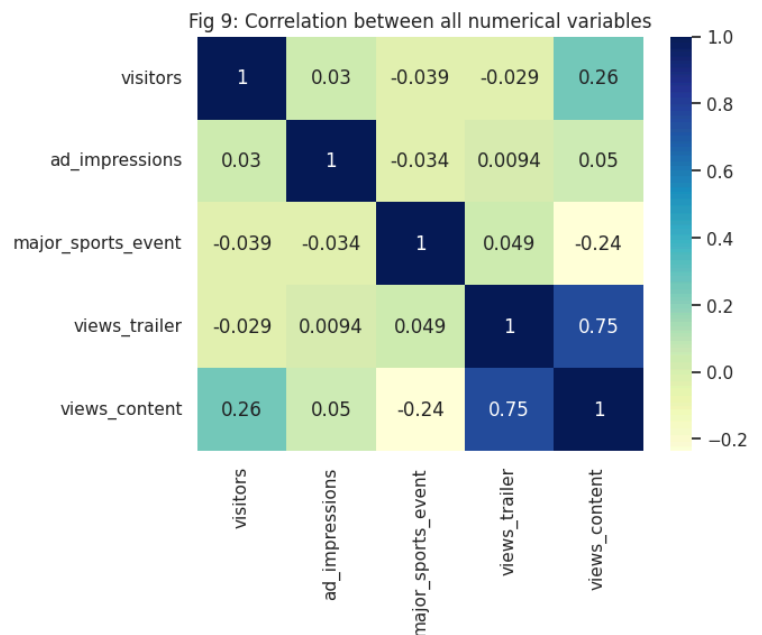


Bivariate analysis

Correlation between all numerical variables

Observation:

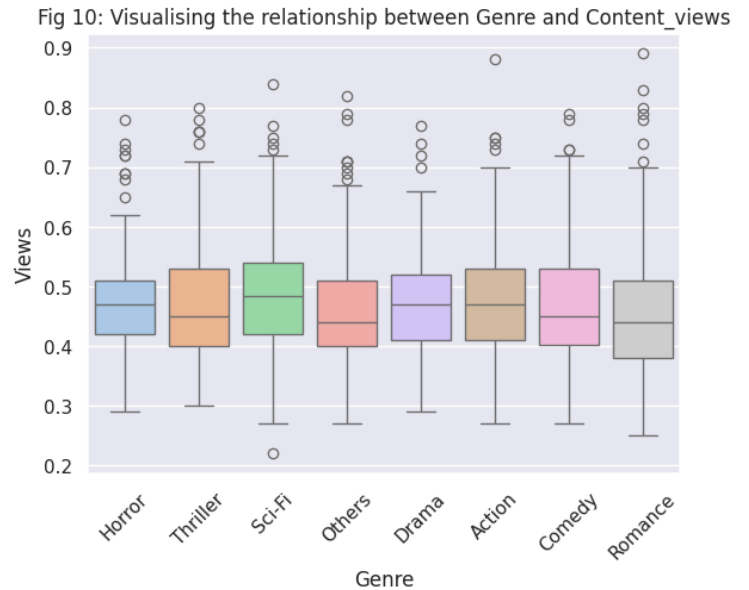
1. Positive correlation can be observed between views content and views trailer.
2. Positive effect of visitor towards the content view can be observed.
3. A negative correlation can be observed between major sports event and content views



Visualising the relationship between Genre and Content_views

Observation:

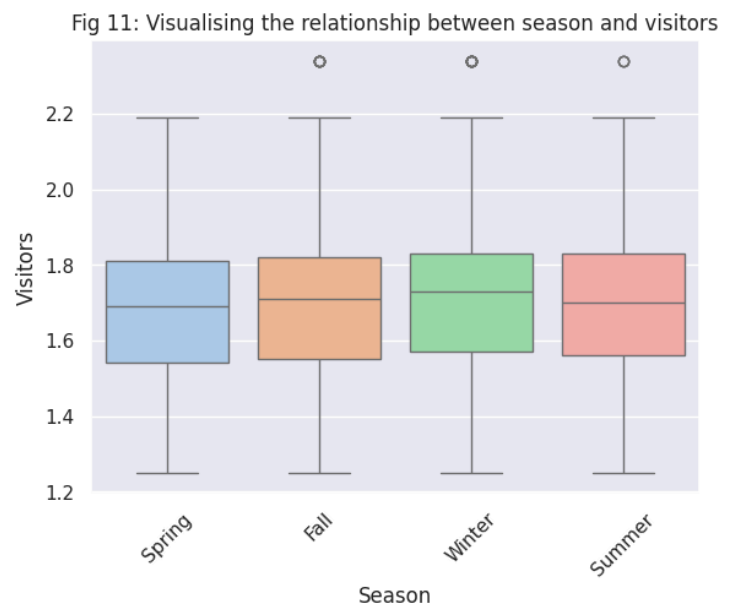
1. Sci-Fi and comedy genres have slightly higher median views compared to others.
2. All genres show a similar range of views(0.3 to 0.7), but there are many outliers, especially above 0.7.



Visualising the relationship between season and visitors

Observation:

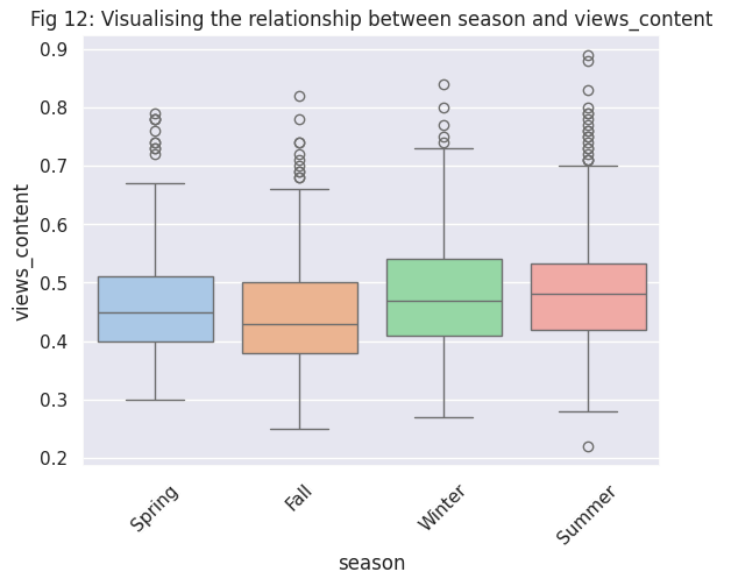
1. There are four seasons and all seasons there is not much difference.



Visualising the relationship between Season and views_content

Observation:

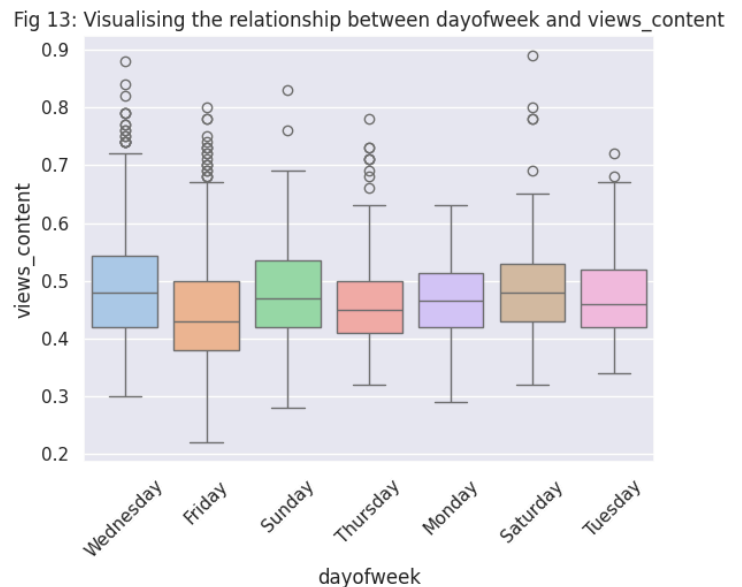
1. There are outliers present for all the season.
2. Average content views for all the seasons looks in between 0.43 to 0.5 million.
3. In the winter and summer mean content views increase than the fall and spring season.



Visualising the relationship between Dayofweeks and views_content

Observation:

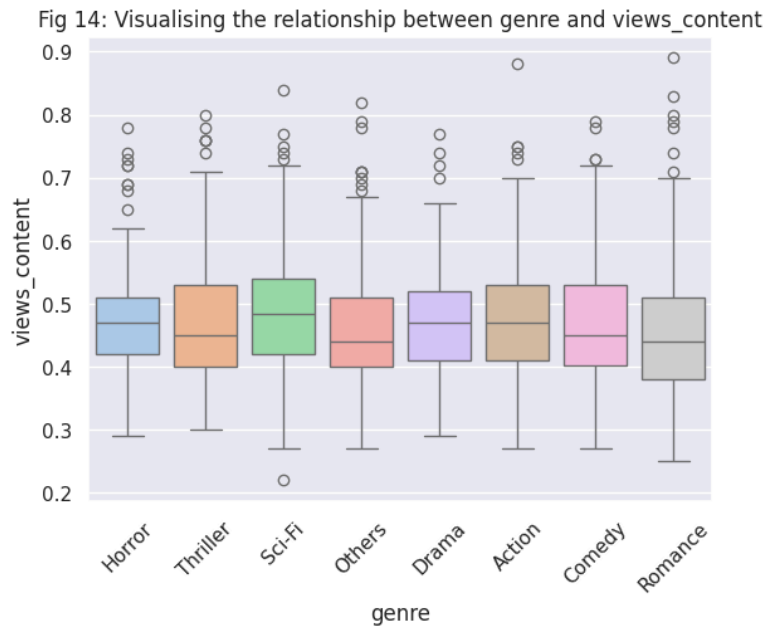
1. There are outliers present for almost all the days of week.
2. Wednesday and Saturday have higher mean values than the other week days.



Visualising the relationship between Genre and views_content

Observation:

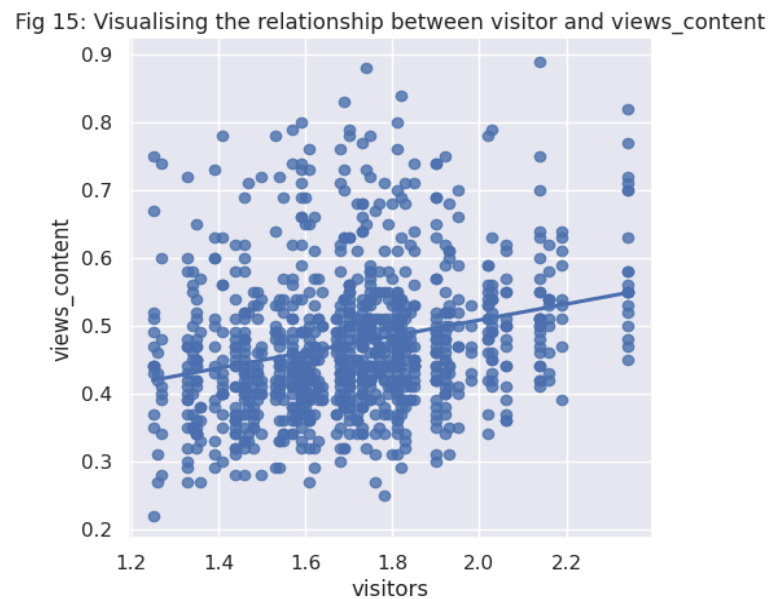
1. There are outliers present for almost all the Genres.
2. Mean value lies between 4 to 5 approximately.



Visualising the relationship between visitor and views_content

Observation:

1. There is a positive correlation between views_content and the visitors.
2. Although no clear pattern, still positive correlation is observed.

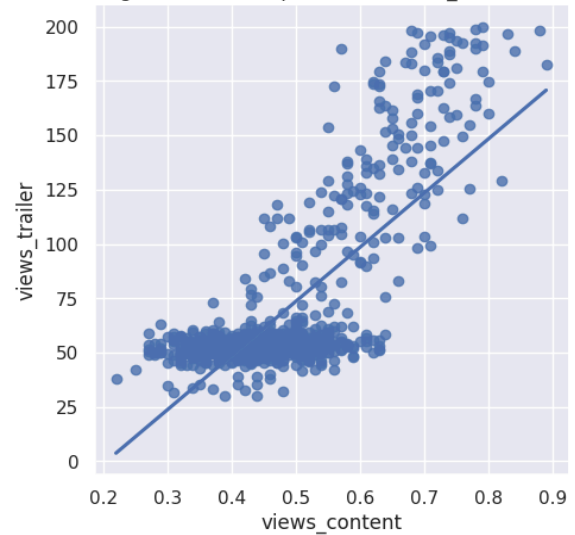


Visualising the relationship between views_content and views_trailer

Observation:

1. The views_content and views_trailer have positive correlation.

Fig 16: Visualising the relationship between views_content and views_trailer



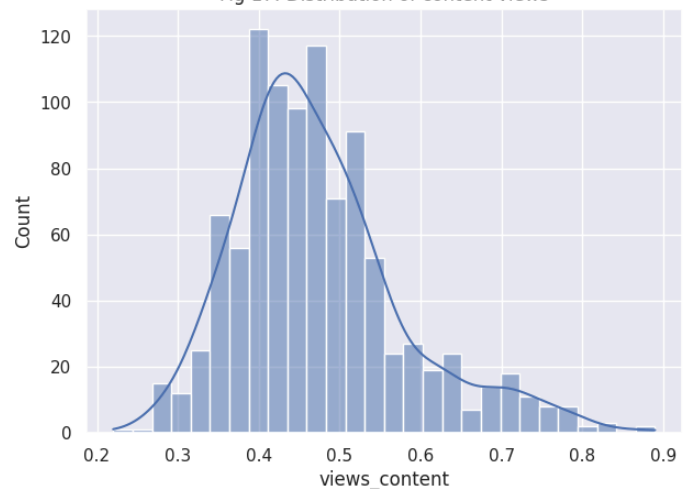
Key Questions

1. What does the distribution of content views look like?

Observation:

1. Little right skewed but looks normally distributed data.
2. Mean value between 4 to 5.

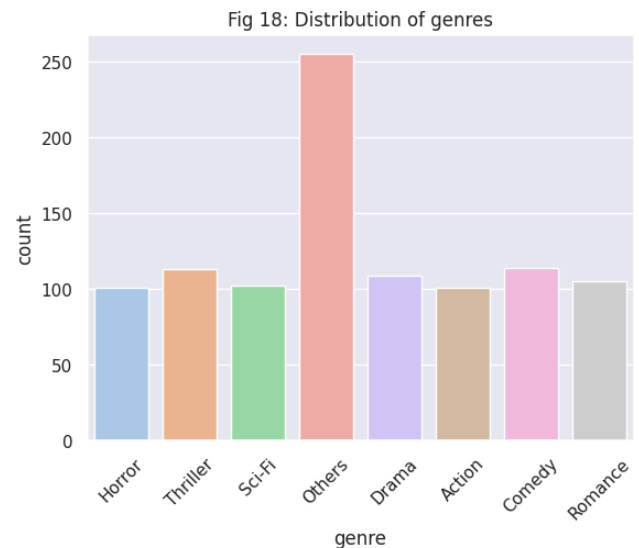
Fig 17: Distribution of content views



2. What does the distribution of genres look like?

Observations:

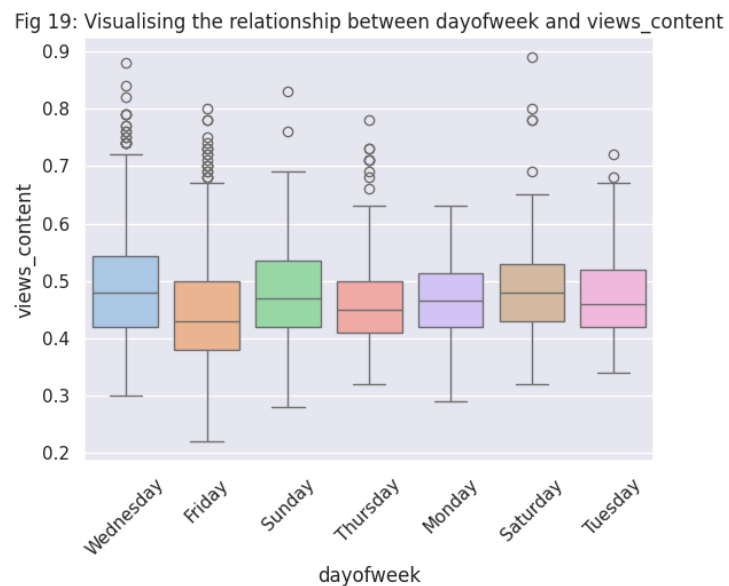
- The count plot reveals that Others is the most popular genre, followed by Comedy and Thriller.



3. The day of the week on which content is released generally plays a key role in the viewership. How does the viewership vary with the day of release?

Observation:

1. There are outliers present for almost all the days of week.
2. Wednesday and Saturday have higher mean values than the other week days.

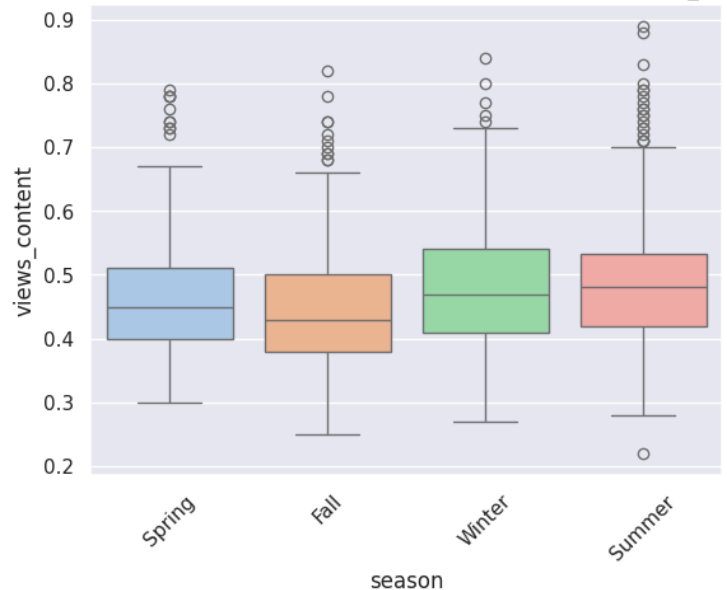


4. How does the viewership vary with the season of release?

Observation:

1. There are outliers present for all the season.
2. Average content views for all the seasons looks in between 0.43 to 0.5 million.
3. In the winter and summer mean content views increase than the fall and spring season.

Fig 20: Visualising the relationship between season and views_content

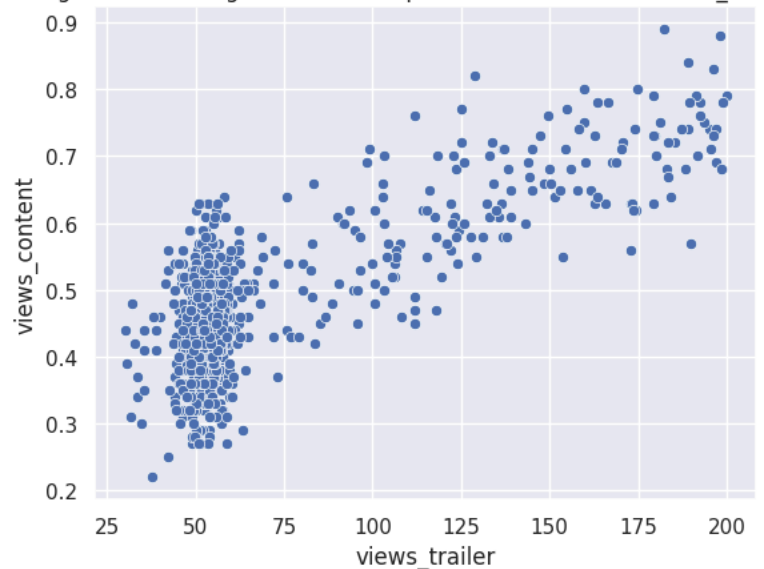


5. What is the correlation between trailer views and content views?

Observations:

1. When trailer views increase, content views also increase.
2. More data points are crowded between 40 to 70 trailer views.
3. High trailer views (above 100) lead to high and stable content views.

Fig 21: Visualising the relationship between visitor and views_content



Data Pre-Processing

Data Preprocessing includes the data for modelling by:

- **Checking for duplicates.**
- **Checking for missing values.**
- **Detecting outliers in content views.**
- **Feature engineering.**
- **Data preparation for modeling.**

Observations:

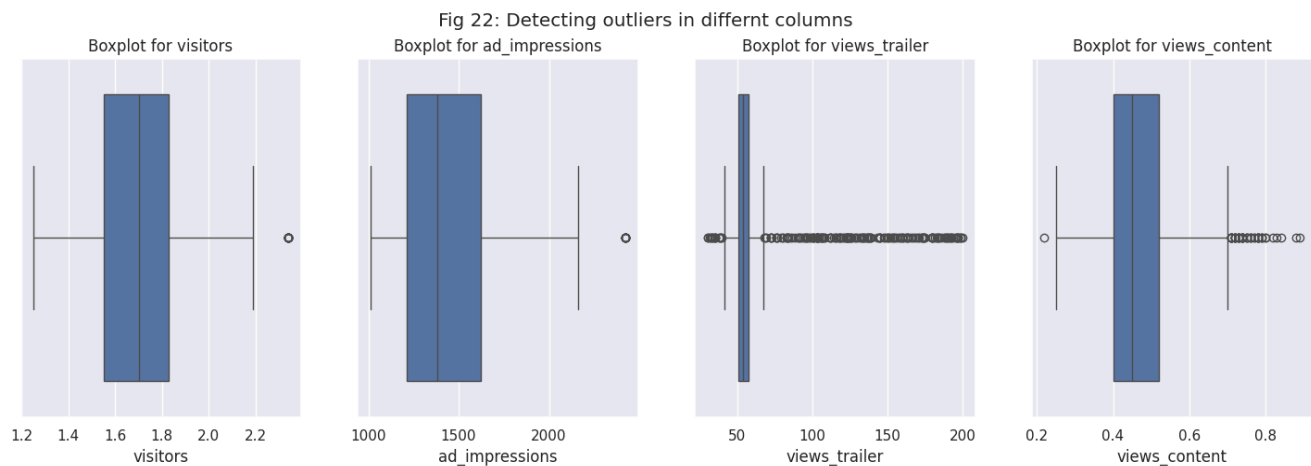
1. Checking for duplicates.

- Using **duplicated()** and **value_counts()** it is found that there are no duplicates in the given dataset.

2. Checking for missing values.

- Using **isnull()** and **sum()** it is found that no missing values are found in the given dataset.

3. Detecting outliers in content views.



Observations:

- There are outliers especially higher in views trailer & views_content but after data analysis we can say they are valid observations.
- So, we will not treat outliers and leave as it is in the dataset and process further and they are Meaningful Outliers.
- These outliers represent valid scenarios because OTT platforms, certain movies or shows might genuinely have significantly higher trailer views or content views due to their popularity, star cast, marketing, etc. so, we cannot remove/treat them.
- Outliers represent valid scenarios, for example, a blockbuster movie's high trailers and views; they should not be removed or treated. These outliers are informative and provide insights into what drives high viewership.

4. Feature engineering.

- For the column major_sports_event, replace the 1 values with 'yes' and 0 values with 'no'.
- Converted the input attributes for float type for modelling.

5. Data preparation for modeling.

- splitting the data into the dependent(y) and independent variables(X).
 - views_content as a dependent variable(y).
 - All the remaining columns except view_content as independent variable(X).
- Creating the dummy variables using **get_dummies()** for the columns(major_sports_event, genre, dayofweek, season) having the object and category.
- **add_constants()** used to add constant terms to the dataset.
- Splitting the data in a 70:30 ratio for train and test data using **train_test_split()**.
- The number of rows in train data is 700.
- The number of rows in test data is 300.

Model Building- Linear Regression

Make the linear model using statsmodels OLS and print the model summary.

```

=====
                        OLS Regression Results
=====
Dep. Variable:          views_content      R-squared:                0.792
Model:                  OLS               Adj. R-squared:          0.785
Method:                 Least Squares      F-statistic:            129.0
Date:                   Thu, 03 Jul 2025   Prob (F-statistic):     1.32e-215
Time:                   07:15:47          Log-Likelihood:         1124.6
No. Observations:       700              AIC:                   -2207.
Df Residuals:           679              BIC:                   -2112.
Df Model:               20
Covariance Type:        nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	0.0602	0.019	3.235	0.001	0.024	0.097
visitors	0.1295	0.008	16.398	0.000	0.114	0.145
ad_impressions	3.623e-06	6.58e-06	0.551	0.582	-9.3e-06	1.65e-05
views_trailer	0.0023	5.52e-05	42.193	0.000	0.002	0.002
major_sports_event_yes	-0.0603	0.004	-15.284	0.000	-0.068	-0.053
genre_Comedy	0.0094	0.008	1.172	0.241	-0.006	0.025
genre_Drama	0.0126	0.008	1.554	0.121	-0.003	0.029
genre_Horror	0.0099	0.008	1.207	0.228	-0.006	0.026
genre_Others	0.0063	0.007	0.897	0.370	-0.008	0.020
genre_Romance	0.0006	0.008	0.065	0.948	-0.016	0.017
genre_Sci-Fi	0.0131	0.008	1.599	0.110	-0.003	0.029
genre_Thriller	0.0087	0.008	1.079	0.281	-0.007	0.025
dayofweek_Monday	0.0337	0.012	2.848	0.005	0.010	0.057
dayofweek_Saturday	0.0579	0.007	8.094	0.000	0.044	0.072
dayofweek_Sunday	0.0363	0.008	4.639	0.000	0.021	0.052
dayofweek_Thursday	0.0173	0.007	2.558	0.011	0.004	0.031
dayofweek_Tuesday	0.0228	0.014	1.665	0.096	-0.004	0.050
dayofweek_Wednesday	0.0474	0.004	10.549	0.000	0.039	0.056
season_Spring	0.0226	0.005	4.224	0.000	0.012	0.033
season_Summer	0.0442	0.005	8.111	0.000	0.034	0.055
season_Winter	0.0272	0.005	5.096	0.000	0.017	0.038

```

=====
Omnibus:                 3.850      Durbin-Watson:           2.004
Prob(Omnibus):           0.146      Jarque-Bera (JB):         3.722
Skew:                    0.143      Prob(JB):                 0.156
Kurtosis:                 3.215      Cond. No.                  1.67e+04
=====

```

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 1.67e+04. This might indicate that there are strong multicollinearity or other numerical problems.

Interpreting the Regression Results:

1. **Adjusted. R-squared:** It reflects the fit of the model.
 - Adjusted R-squared values generally range from 0 to 1, where a higher value generally indicates a better fit, assuming certain conditions are met.
 - In our case, the value for adj. R-squared is **0.785**, which is good.
2. **const coefficient:** It is the Y-intercept.
 - It means that if all the predictor variable coefficients are zero, then the expected output (i.e., Y) would be equal to the const coefficient.
 - In our case, the value for const coefficient is **0.0602**.
3. **Coefficient of a predictor variable:** It represents the change in the output Y due to a change in the predictor variable (everything else held constant).
 - In our case, the coefficient of duration is **0.1295**.

Display model coefficients with column names

	Column_Names	Coefficient
0	const	0.060157
1	visitors	0.129451
2	ad_impressions	0.000004
3	views_trailer	0.002330
4	major_sports_event_yes	-0.060326
5	genre_Comedy	0.009352
6	genre_Drama	0.012625
7	genre_Horror	0.009862
8	genre_Others	0.006325
9	genre_Romance	0.000551
10	genre_Sci-Fi	0.013143
11	genre_Thriller	0.008708
12	dayofweek_Monday	0.033662
13	dayofweek_Saturday	0.057887
14	dayofweek_Sunday	0.036321
15	dayofweek_Thursday	0.017289
16	dayofweek_Tuesday	0.022837
17	dayofweek_Wednesday	0.047376
18	season_Spring	0.022602
19	season_Summer	0.044203
20	season_Winter	0.027161

Model Performance Check

Testing the model performance on train and test data

Train Performance				
RMSE	MAE	MAPE	R2	Adj_R2
0.04853	0.038197	8.55644	0.791616	0.785162

Test Performance				
RMSE	MAE	MAPE	R2	Adj_R2
0.050603	0.040782	9.030464	0.766447	0.748804

Observation on 1st Model:

1. The training R2 is 0.791616, so the model is not underfitting
2. The train and test RMSE and MAE are comparable, so the model is not overfitting either
3. MAE suggests that the model can predict content views within a mean error of 0.04 on the test data
4. MAPE of 9.03 on the test data means that we are able to predict within 9.03% of the content views.

Checking Linear Regression Assumptions

We will be checking the following Linear Regression assumptions:

1. No Multicollinearity
2. Linearity of variables
3. Independence of error terms
4. Normality of error terms
5. No Heteroscedasticity

Test For Multicollinearity

feature	VIF	
0	const	99.679317
1	visitors	1.027837
2	ad_impressions	1.029390
3	views_trailer	1.023551
4	major_sports_event_yes	1.065689
5	genre_Comedy	1.917635
6	genre_Drama	1.926699
7	genre_Horror	1.904460
8	genre_Others	2.573779
9	genre_Romance	1.753525
10	genre_Sci-Fi	1.863473
11	genre_Thriller	1.921001
12	dayofweek_Monday	1.063551
13	dayofweek_Saturday	1.155744
14	dayofweek_Sunday	1.150409
15	dayofweek_Thursday	1.169870
16	dayofweek_Tuesday	1.062793
17	dayofweek_Wednesday	1.315231
18	season_Spring	1.541591
19	season_Summer	1.568240
20	season_Winter	1.570338

Observation on VIF values:

1. After computing the variance inflation factor it is seen that "genre_Others" has a VIF value more than 2.0.
2. Other variables are having the variance inflation factor less than 2.0 value.
3. We can conclude that no value is greater than 5 , stating there is no multicollinearity .

Dropping high p-value variables

The linear model using statsmodels OLS and prints the model summary after removing p-value.

```

=====
                        OLS Regression Results
=====
Dep. Variable:          views_content      R-squared:                0.789
Model:                  OLS               Adj. R-squared:          0.786
Method:                 Least Squares     F-statistic:            233.8
Date:                   Thu, 03 Jul 2025   Prob (F-statistic):      7.03e-224
Time:                   09:30:33          Log-Likelihood:         1120.2
No. Observations:       700              AIC:                   -2216.
Df Residuals:           688              BIC:                   -2162.
Df Model:                11
Covariance Type:        nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	0.0747	0.015	5.110	0.000	0.046	0.103
visitors	0.1291	0.008	16.440	0.000	0.114	0.145
views_trailer	0.0023	5.5e-05	42.414	0.000	0.002	0.002
major_sports_event_yes	-0.0606	0.004	-15.611	0.000	-0.068	-0.053
dayofweek_Monday	0.0321	0.012	2.731	0.006	0.009	0.055
dayofweek_Saturday	0.0570	0.007	8.042	0.000	0.043	0.071
dayofweek_Sunday	0.0344	0.008	4.456	0.000	0.019	0.050
dayofweek_Thursday	0.0154	0.007	2.307	0.021	0.002	0.029
dayofweek_Wednesday	0.0465	0.004	10.532	0.000	0.038	0.055
season_Spring	0.0226	0.005	4.259	0.000	0.012	0.033
season_Summer	0.0434	0.005	8.112	0.000	0.033	0.054
season_Winter	0.0282	0.005	5.362	0.000	0.018	0.039

```

=====
Omnibus:                 3.254      Durbin-Watson:           1.996
Prob(Omnibus):            0.196      Jarque-Bera (JB):         3.077
Skew:                     0.139      Prob(JB):                 0.215
Kurtosis:                 3.168      Cond. No.                  662.
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Testing the model performance on train and test data

Train Performance				
RMSE	MAE	MAPE	R2	Adj_R2
0.048841	0.038385	8.595246	0.788937	0.785251

Test Performance				
RMSE	MAE	MAPE	R2	Adj_R2
0.051109	0.041299	9.177097	0.761753	0.751792

Observations

1. Now no feature has p-value greater than 0.05, so we'll consider the features in *x_train1* as the final set of predictor variables and *olsmodel2* as the final model to move forward with
2. Now adjusted R-squared is 0.786, i.e., our model is able to explain ~78% of the variance
3. This shows that the variables we dropped were not affecting the model
4. RMSE and MAE values are comparable for train and test sets, indicating that the model is not overfitting.

Test for Normality and Independence

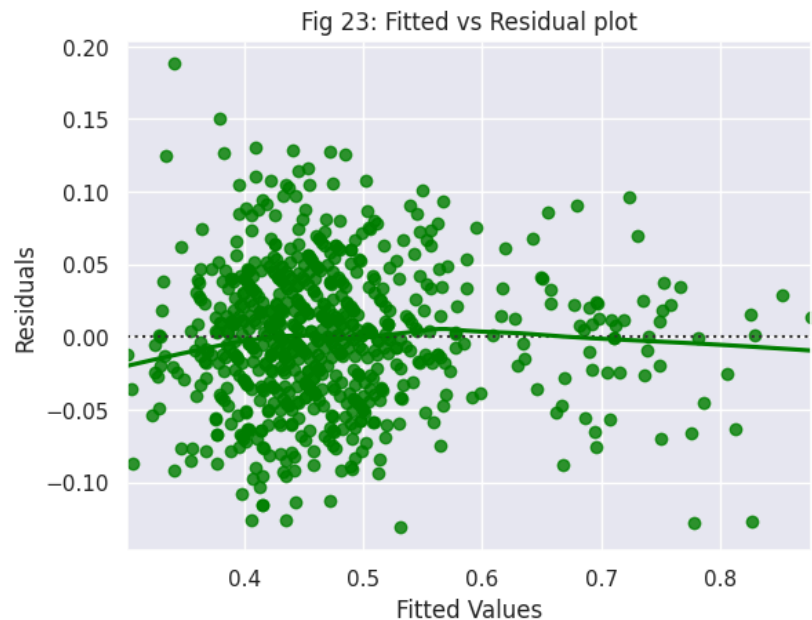
- Created Dataframe with actual, fitted and residual values.

	Actual Values	Fitted Values	Residuals
731	0.40	0.445434	-0.045434
716	0.70	0.677403	0.022597
640	0.42	0.433999	-0.013999
804	0.55	0.562030	-0.012030
737	0.59	0.547786	0.042214

Plotted the graph between fitted_values and residuals

Observations

- The scatter plot shows the distribution of residuals (errors) vs fitted values (predicted values).
- If there exists any pattern in this plot, we consider it as signs of non-linearity in the data and a pattern means that the model doesn't capture non-linear effects.
- We see no pattern in the plot above. Hence, the assumptions of linearity and independence are satisfied.**

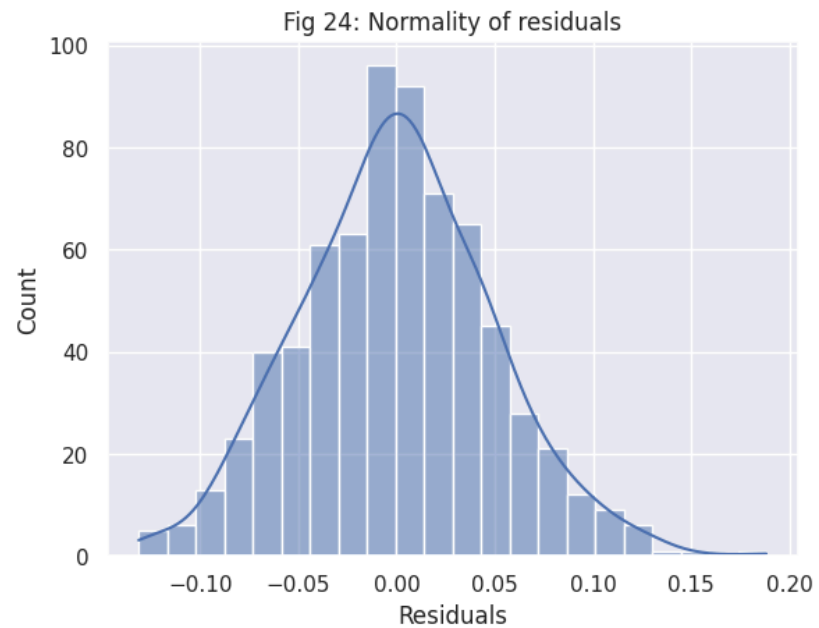


TEST FOR NORMALITY

Checked the normality of residuals using histplot()

Observations

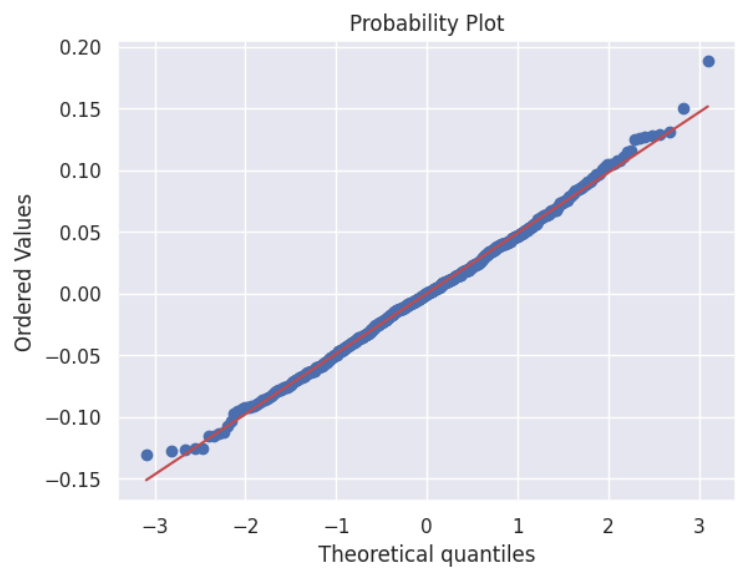
- The histogram of residuals almost has a bell shaped structure.



Checked the normality of residuals using Q-Q plot

Observations

- The residuals almost follow a straight line except for the tails.



Checked the normality of residuals using the Shapiro-Wilk test.

- The p-value(0.31085896470099983) is greater than 0.05, so the residuals are normally distributed as per the Shapiro-Wilk test.

Observations for Normality

- Since p-value < 0.05 , the residuals are not normal as per the Shapiro-Wilk test.
- Strictly speaking, the residuals are not normal. However, as an approximation, we can accept this distribution as close to being normal.
- **So, the assumption is satisfied.**

Test for Homoscedasticity

Observations

- Since p-value > 0.05 , the residuals are homoscedastic.
- **So, the assumption is satisfied.**

All the assumptions of linear regression are satisfied. Let's rebuild our final model, check its performance, and draw inferences from it.

Predictions on test data

	Actual	Predicted
983	0.43	0.434802
194	0.51	0.500314
314	0.48	0.430257
429	0.41	0.492544
267	0.41	0.487034
746	0.68	0.680000
186	0.62	0.595078
964	0.48	0.503909
676	0.42	0.490313
320	0.58	0.560155

- We can observe here that our model has returned pretty good prediction results, and the actual and predicted values are comparable.

Model performance evaluation

OLS Regression Results						
Dep. Variable:	views_content	R-squared:	0.789			
Model:	OLS	Adj. R-squared:	0.786			
Method:	Least Squares	F-statistic:	233.8			
Date:	Thu, 03 Jul 2025	Prob (F-statistic):	7.03e-224			
Time:	09:30:33	Log-Likelihood:	1120.2			
No. Observations:	700	AIC:	-2216.			
Df Residuals:	688	BIC:	-2162.			
Df Model:	11					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0.0747	0.015	5.110	0.000	0.046	0.103
visitors	0.1291	0.008	16.440	0.000	0.114	0.145
views_trailer	0.0023	5.5e-05	42.414	0.000	0.002	0.002
major_sports_event_yes	-0.0606	0.004	-15.611	0.000	-0.068	-0.053
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Omnibus:	3.254	Durbin-Watson:	1.996			
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Kurtosis:	3.168	Cond. No.	662.			
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Train Performance				
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Test Performance				
RMSE	MAE	MAPE	R2	Adj_R2
0.051109	0.041299	9.177097	0.761753	0.751792

Observation:

- We can see there is not much significant difference between Training and Test Data, so we can say our model is good.
- RMSE value not much changed from Training/Test result.
- MAE value also not having much differences in Training/Test result.
- R-Square just decreased in Testing as compared to training but this is marginal and can be accepted.
- Adj. R-Square also decreased in test results but this change is also valid and acceptable.

Actionable Insights & Recommendations

- The model's R-squared value is approximately 0.788, and the adjusted R-squared is 0.785, indicating that the model can explain about 79% of the variance in the data. This is quite satisfactory.
- An increase of one unit in the 'visitors' variable results in a 0.1291 unit increase in content viewership, with all other variables held constant.
- Releasing content on Saturdays and Wednesdays will boost viewership, provided no major sports events occur on those days.
- Releasing content during the summer season can enhance viewership
- To improve content viewership, it is recommended to avoid releasing content on days when major sports events are happening.
- In Genre "Other" categories have the highest counts followed by the Comedy, Thriller, Drama etc.
- A major sports event will lead to a 0.0606 unit decrease in content viewership.
- The summer season can result in a 0.0434 unit increase in content viewership.